BIOL 425 Final Report: BosR DNA Binding Coordinates RpoS-Dependent and Independent Gene Regulation During Mammalian Host Adaptation in *Borrelia burgdorferi*

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1 Introduction

1.1 Background

Borrelia burgdorferi is the causing agent of Lyme disease so it must adapt its gene expression as it transitions between tick vectors and mammalian hosts. BosR is one important transcriptional regulator involved in this process since it is a ferric uptake regulator (FUR) family protein that controls expression of genes that are associated with mammalian host adaptation and virulence. Many previous studies heavily focused on

BosR's role in activating the alternative sigma factor RpoS and its role in regulating oxidative stress response genes, however, the broader significance of BosR to RpoS-dependent and RpoS-independent transcriptional regulation remained unclear. In this paper, Grassmann et al. (2025) investigated how BosR DNA-binding activity influences global gene expression in B. burgdorferi under both in vitro and mammalian host adapted conditions using transcriptomic analysis.

The reanalysis performed allowed us to focus on comparing transcriptomic differences between wild-type (WT) and ΔbosR strains that are cultivated under in vitro and dialysis membrane chamber (DMC) conditions. The analysis allows us to evaluate the changes in gene expression that are associated with BosR disruption and mammalian host adaptation using RNA-seq derived expression data. This work aims to identify genes that are significantly altered between conditions and allows us to visualize these differences using fold-change analysis and volcano plots.

1.2 Biological hypothesis

BosR regulates transcription in Borrelia burgdorferi through both RpoS-dependent and RpoS-independent mechanisms and when BosR DNA binding is disrupted it significantly alters global gene expression during mammalian host adaptation.

1.3 Significance

It is very important to clearly understand how B. burgdorferi adapts to mammalian hosts in order to understand Lyme disease pathogenesis. BosR function as a transcriptional regulator which coordinates expression of virulence associated genes, host adaptation pathways, and genes that are involved in bacterial survival.

It is important to be able to determine whether BosR acts only through RpoS or independently of RpoS since it helps clarify the organization of transcriptional networks in B. burgdorferi. It has always been assumed that BosR functions as an oxidative stress regulator so these findings challenge these assumptions since the results suggest that BosR acts as a global regulatory protein that is involved in mammalian phase gene expression.

This work could contribute to future efforts that can help identify therapeutic targets for Lyme disease since it helps improve the understanding of the molecular mechanisms that control bacterial adaptation and virulence.

2 Materials and Methods

2.1 Samples

The study used *Borrelia burgdorferi* strain B31 5A4 as the wild-type strain. The mutant strains had a ΔbosR mutant that lacked the *bosR* gene and a DNA-binding-defective BosR-

R39A mutant. The ΔrpoS strain was also used in order to compare it with the RpoS-dependent transcriptional responses.

The samples were analyzed under two conditions:

1. In vitro cultivated spirochetes grown in BSK-II medium after temperature shift.
2. Mammalian host-adapted spirochetes cultivated in dialysis membrane chambers (DMCs) implanted in rats.

Biological replicates were also collected for RNA sequencing analysis where the wild-type strain served as the experimental control for comparison with mutant strains.

2.2 Experimental procedure

RNA-seq expression datasets from Grassmann et al. (2025) were analyzed in R using multiple bioinformatics and visualization packages like tidyverse, readxl, EnhancedVolcano, ggplot2, ggrepel, and preprocessCore. Then Gene expression tables were imported from the Excel files that contained transcriptomic comparisons between WT, ΔbosR, and BosR-R39A mutant strains under in vitro and DMC conditions.

The data was then reshaped into long format using the pivot_longer() function in order to facilitate visualization and statistical analysis. Histograms were generated to examine the distribution of normalized RFPKM expression values across biological replicates and experimental treatments. Then WT in vitro samples and WT DMC samples were merged by GeneID in order to be able to compare gene expression between the environmental conditions.

Quantile normalization was performed using the normalize.quantiles() function from the preprocessCore package in order to reduce technical variation across samples where the normalized data was then log2-transformed prior to analysis. Mean expression values, fold change, and log2 fold changes for in vitro and DMC conditions were calculated for each gene in order to to quantify transcriptional differences between conditions. Volcano plots and histograms were generated to visualize differential gene expression patterns between samples.

2.3 Statistical methods

Statistical analyses were conducted in R in order to find the differential gene expression between experimental conditions. Mean normalized expression values were calculated across biological replicates for each condition, and fold changes were computed as the ratio of DMC expression to in vitro expression. Log2 transformation of fold change values was used to stabilize variance in order to improve interpretability of upregulated and downregulated genes.

For each gene, an independent two sample t-test was performed in order to compare normalized expression values between in vitro and DMC conditions. Genes with zero variance across replicates were excluded from statistical testing in order to avoid invalid calculations and p-values were adjusted for multiple hypothesis testing using the Benjamini–Hochberg (BH) false discovery rate correction method in order to get the adjusted p-values (padj). Exploratory data analysis like histogram distributions, scatter plots, and volcano plots were performed in order to visualize the magnitude and significance of transcriptional changes.

3 Exploratory data analysis

Re-analysis of Grassmann et al (2025) Borrelia transcriptome comparisons

Part 1. load libraries; set working directory

```
library(tidyverse)

## Warning: package 'ggplot2' was built under R version 4.5.2

## Warning: package 'readr' was built under R version 4.5.2

## Warning: package 'purrr' was built under R version 4.5.2

## Warning: package 'dplyr' was built under R version 4.5.2

## ── Attaching core tidyverse packages ──────────────────────── tidyverse
2.0.0 ──
## ✓ dplyr     1.2.1     ✓ readr     2.2.0
## ✓ forcats   1.0.1     ✓ stringr   1.6.0
## ✓ ggplot2   4.0.3     ✓ tibble    3.3.1
## ✓ lubridate 1.9.5     ✓ tidyr     1.3.2
## ✓ purrr     1.2.2
## ── Conflicts ──────────────────────────────────────────
tidyverse_conflicts() ──
## ✕ dplyr::filter() masks stats::filter()
## ✕ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors

library(readxl)
library(dplyr)
library(EnhancedVolcano)

## Warning: package 'EnhancedVolcano' was built under R version 4.5.2

## Loading required package: ggrepel

## Warning: package 'ggrepel' was built under R version 4.5.2

library(ggrepel)
library(shiny)
```

```
## Warning: package 'shiny' was built under R version 4.5.2

library(bslib)

##
## Attaching package: 'bslib'
##
## The following object is masked from 'package:utils':
##
##     page

library(modeldata)

##
## Attaching package: 'modeldata'
##
## The following object is masked from 'package:datasets':
##
##     penguins

library(DataExplorer)

## Warning: package 'DataExplorer' was built under R version 4.5.2

library(plotly)

##
## Attaching package: 'plotly'
##
## The following object is masked from 'package:ggplot2':
##
##     last_plot
##
## The following object is masked from 'package:stats':
##
##     filter
##
## The following object is masked from 'package:graphics':
##
##     layout

library(ggplot2)
library(BiocManager)

## Bioconductor version '3.22' is out-of-date; the current release version
'3.23'
##   is available with R version '4.6'; see https://bioconductor.org/install

setwd("~/QiuLab-work/")
```

Part 2. Read data table, reshape into long tables

```
x <- read_excel("mmi70036-sup-0002-tables2.xlsx", sheet = 3)
colnames(x)

##  [1] "GeneID"             "baseMean"           "log2FoldChange"
##  [4] "lfcSE"              "stat"               "pvalue"
##  [7] "padj"               "WT_in_vitro_1"      "WT_in_vitro_2"
## [10] "WT_in_vitro_3"      "ΔbosR _in_vitro_1" "ΔbosR _in_vitro_2"
## [13] "ΔbosR _in_vitro_3" "ΔbosR _in_vitro_4"

x.long <- x |>
    select(1, 8:14) |>
    pivot_longer(2:8, names_to = "treat", values_to = "rfpkm")

x.long <- x.long |>
    mutate(rep = as.numeric(str_remove(treat, "^.*_")))

x.long <- x.long |>
    mutate(treat2 = str_remove(treat, "_\\d$"))
```

Part 3. Plot histogram to show distributions

```
x.long |>
    ggplot(aes(rfpkm+1, fill = as.factor(rep))) +
    geom_histogram(bins = 100) +
    scale_x_log10() +
    facet_wrap(~treat2, nrow=2) +
    theme_bw() +
    labs(
      title = "Figure 1. Distribution of Normalized Gene Expression Values",
      x = "Log10(RFPKM + 1)",
      y = "Gene Count",
      fill = "Replicate")
```

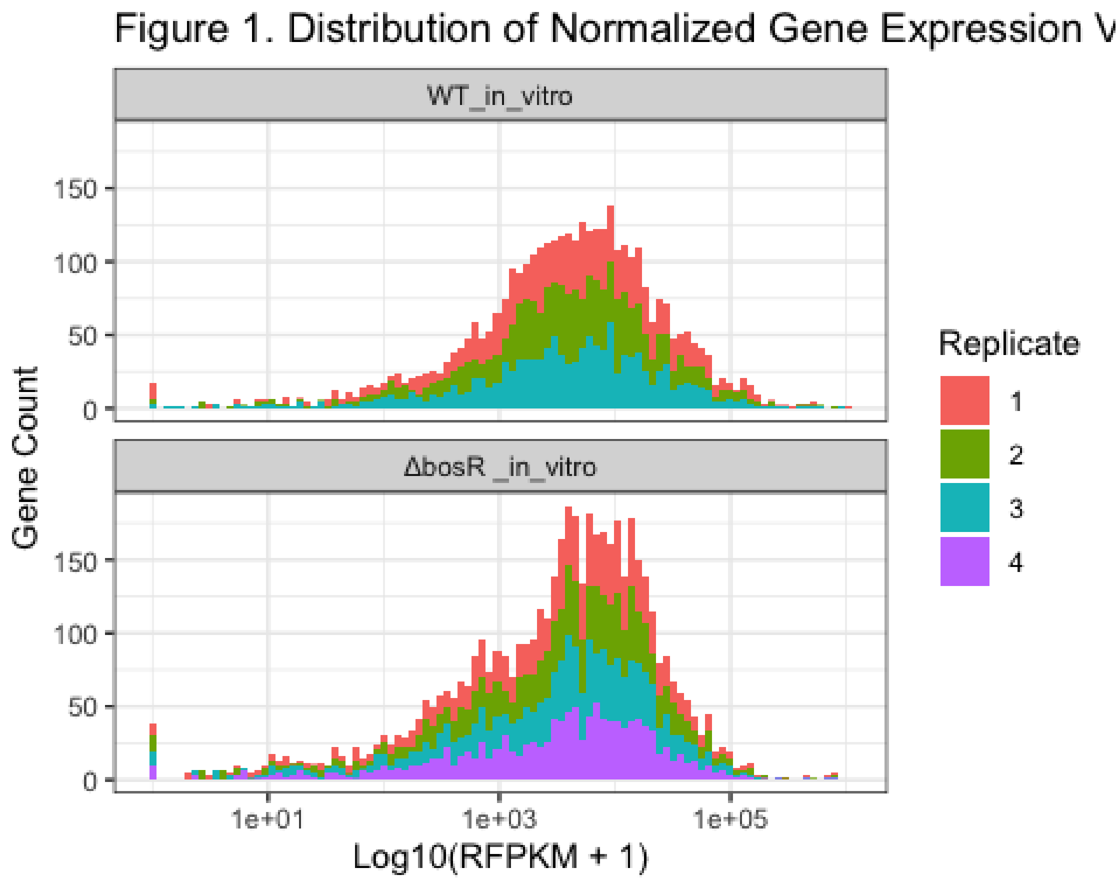

Figure 1. Distribution of normalized RFPKM gene expression values across biological replicates under in vitro and DMC conditions. Expression values were quantile normalized and log2 transformed prior to analysis. Similar distributions across replicates indicate successful normalization and reduced technical variation among samples.

Part 4. read data frame

```
x <- read_excel("mmi70036-sup-0002-tables2.xlsx", sheet = 3)
y <- read_excel("mmi70036-sup-0003-tables3.xlsx", sheet = 3)
```

Part 5. select desired data

```
in_vitro <- x[, c(1, 8, 9, 10)]
colnames(in_vitro) <- c("GeneID", "WT_in_vitro_1", "WT_in_vitro_2",
"WT_in_vitro_3")


DMC <- y[, c(1, 8:13)]
colnames(DMC) <-
c("GeneID","WT_DMC1","WT_DMC2","WT_DMC3","WT_DMC4","WT_DMC5","WT_DMC6")
```

```
WT <- merge(in_vitro, DMC , by = "GeneID")
```

#normilization

```
library(preprocessCore)

## Warning: package 'preprocessCore' was built under R version 4.5.2

normalized_data <- normalize.quantiles(as.matrix(WT[, -1]))

WT <- data.frame(GeneID = WT$GeneID, normalized_data)
colnames(WT)[-1] <- c("WT_in_vitro_1","WT_in_vitro_2","WT_in_vitro_3",
                      "WT_DMC1","WT_DMC2","WT_DMC3",
                      "WT_DMC4","WT_DMC5","WT_DMC6")

WT[, -1] <- log2(WT[, -1] + 1)
```

Part 6. calculate fold change

```
FC <- WT %>%
mutate(in_vitro_mean = rowMeans(select(., 2:4)),
 DMC_mean = rowMeans(select(., 5:10)),
 fold_change = DMC_mean / in_vitro_mean,
 log2FC = log2(fold_change))

FC |>
  ggplot(aes(x = log2FC)) +
  geom_histogram(bins = 100) +
  geom_vline(xintercept = 0, col = "red") +
  theme_bw() +
  labs(
    title = "Figure 2. WT In Vitro vs WT DMC Differential Expression",
    x = "Log2 Fold Change",
    y = "Gene Count")
```

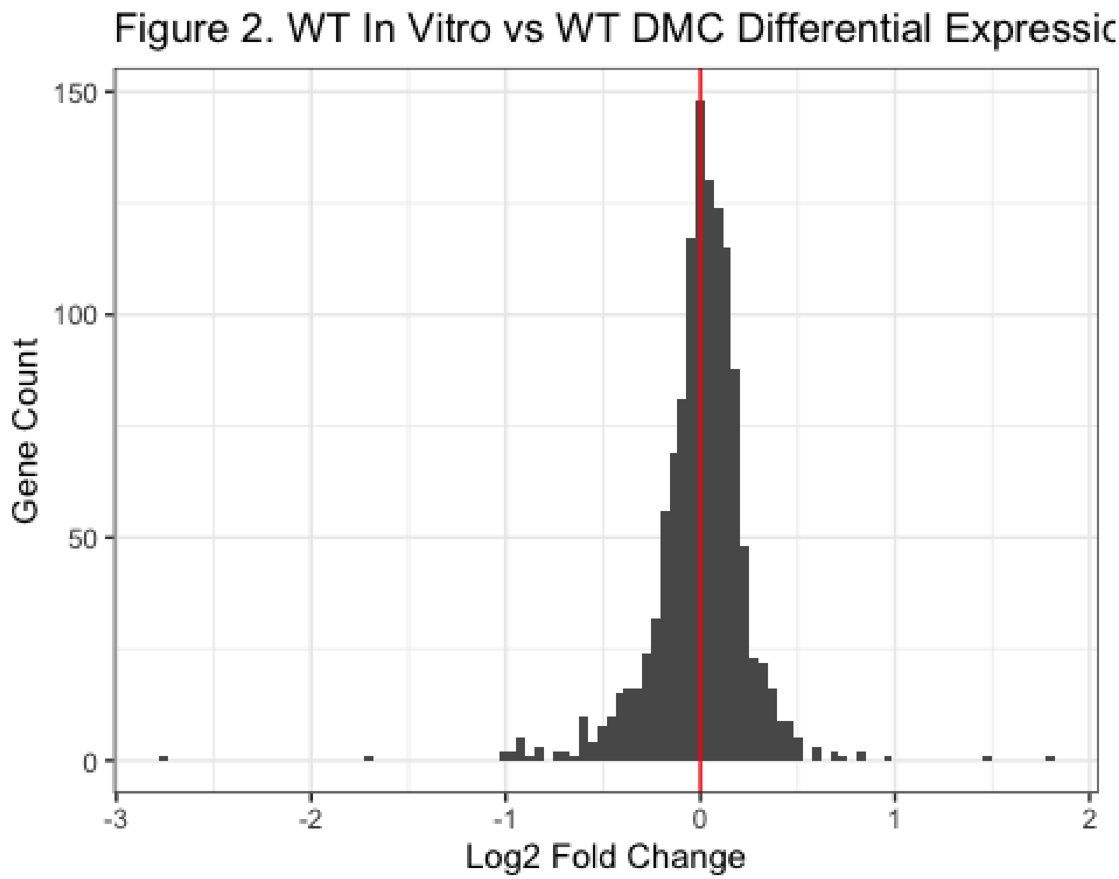

Figure 2. Histogram showing the distribution of log2 fold-change values between WT in vitro and WT DMC conditions. The red vertical line represents no change in gene expression (log2FC = 0). Genes farther from zero exhibit stronger differential expression associated with mammalian host adaptation.

Part 7. Calculate pvalue

```
FC <- FC %>%
  rowwise() %>%
  mutate(
  pvalue = if(sd(c_across(2:4)) == 0 | sd(c_across(5:10)) == 0) NA_real_
    else t.test(c_across(2:4), c_across(5:10))$p.value) %>%
  ungroup() %>%
  mutate(p_adj = p.adjust(pvalue, method = "BH"))

FC_clean <- FC %>%
  filter(is.finite(log2FC), !is.na(pvalue))

# show BB_Q67, methyl-transferase
gene.sel <- FC_clean |> filter(GeneID %in% c('BB_Q67', 'BB_A15', 'BB_B19',
'BB_B24'))
#|> select(in_vitro_mean, DMC_mean, fold_change, pvalue, p_adj)
```

Part 8. plot volcano plot

```
EnhancedVolcano(
  toptable = FC_clean,
  lab = FC_clean$GeneID,
  x = "log2FC",
  y = "pvalue",
  title = "Figure 3. Volcano Plot of WT In Vitro vs WT DMC",
  pCutoff = 0.05,
  FCcutoff = 1,
  pointSize = 2.0,
  labSize = 3.0)

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## ℹ The deprecated feature was likely used in the EnhancedVolcano package.
##   Please report the issue to the authors.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Warning: The `size` argument of `element_line()` is deprecated as of
ggplot2 3.4.0.
## ℹ Please use the `linewidth` argument instead.
## ℹ The deprecated feature was likely used in the EnhancedVolcano package.
##   Please report the issue to the authors.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

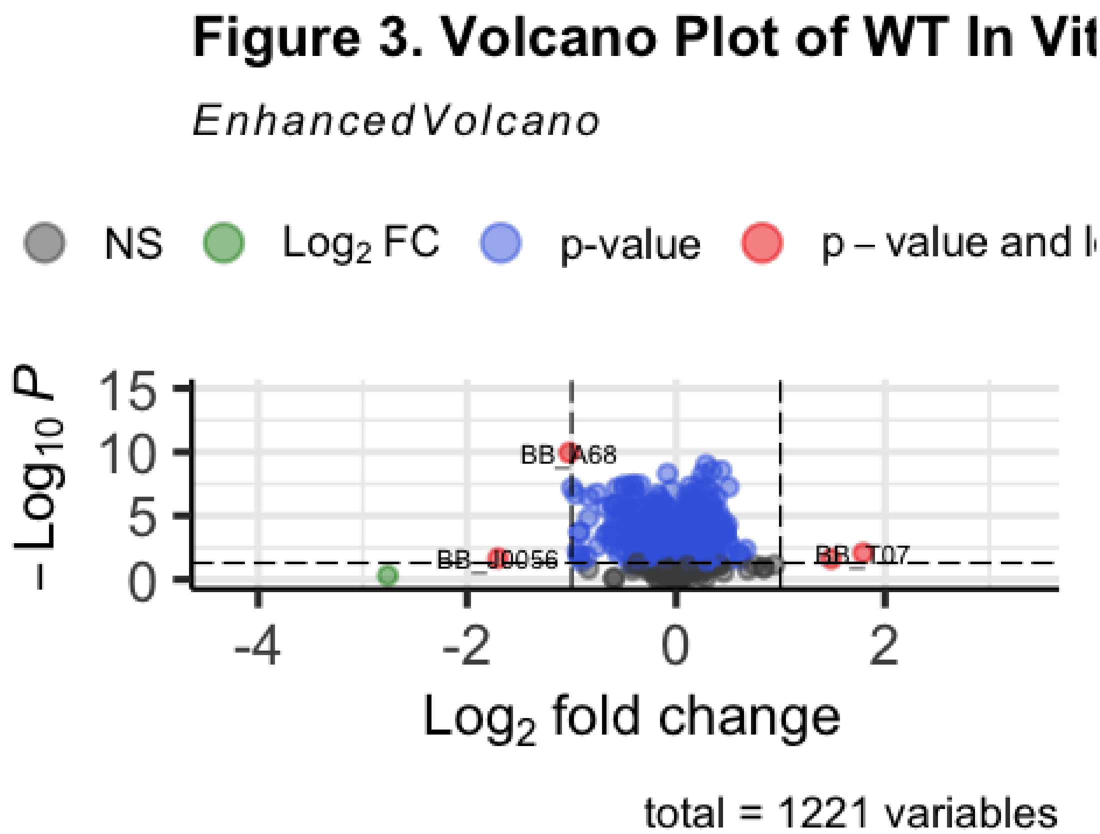

Figure 3. Volcano plot comparing WT in vitro and WT DMC gene expression profiles. Each point represents a gene plotted according to log2 fold change and adjusted p-value. Red points indicate selected genes of interest with statistically significant differential expression.

```
FC_clean |>
  ggplot(aes(log2FC, p_adj, label = GeneID)) +
  geom_point(shape = 1) +
  geom_point(data = gene.sel, col = "red") +
  geom_text_repel(data = gene.sel, col = "red") +
  scale_y_log10() +
  geom_vline(xintercept = 0, linetype = 2) +
  geom_hline(yintercept = 1e-3, col = "red") +
  theme_bw() +
  labs(
    title = "Figure 4. WT In Vitro vs WT DMC Volcano Plot",
    x = "Log2 Fold Change",
    y = "Adjusted p-value"
  )
```

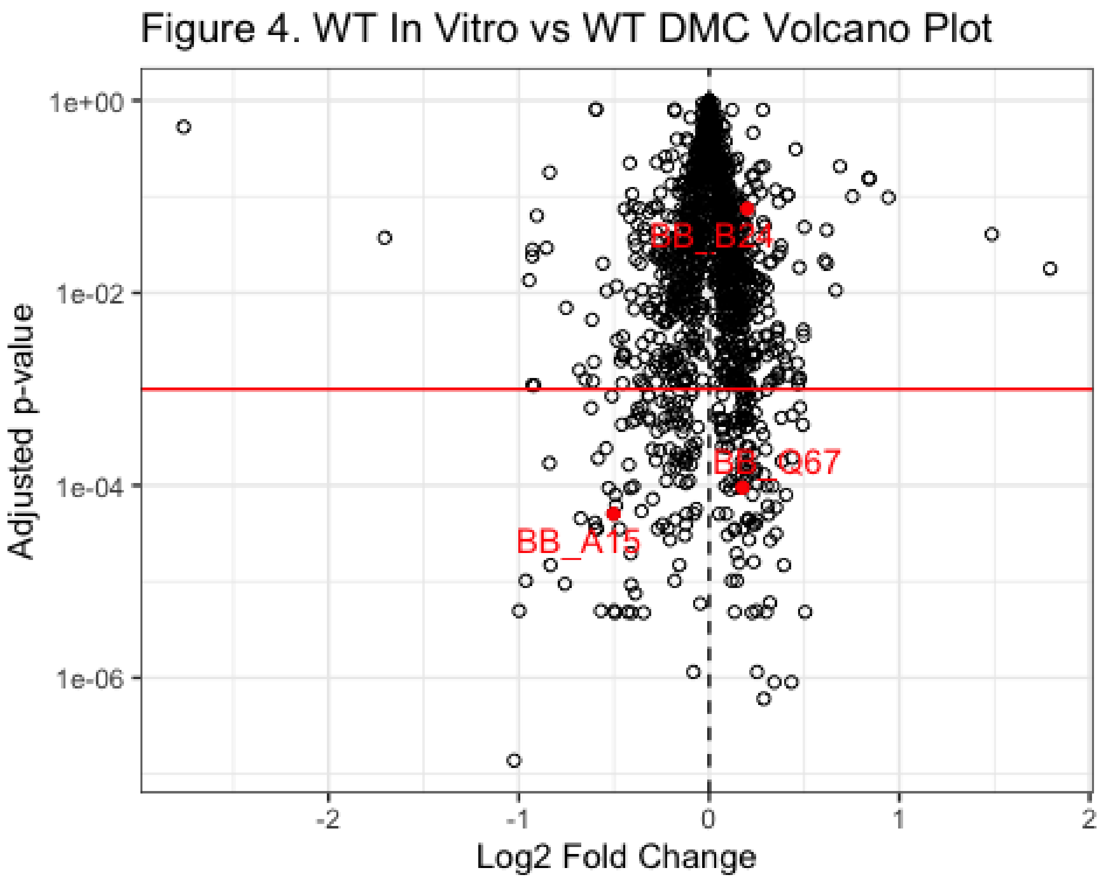

Figure 4. Scatter volcano plot comparing WT in vitro and WT DMC transcriptomes. Genes are plotted according to log2 fold change and adjusted p-values. Highlighted red points represent selected genes of biological interest showing strong differential expression between conditions.

#Table 2 - in vitro

```
WT_vs_bosR_in_vitro <- read_excel("mmi70036-sup-0002-tables2.xlsx", sheet =
3)
colnames(WT_vs_bosR_in_vitro)

##  [1] "GeneID"            "baseMean"          "log2FoldChange"
##  [4] "lfcSE"             "stat"              "pvalue"
##  [7] "padj"              "WT_in_vitro_1"     "WT_in_vitro_2"
## [10] "WT_in_vitro_3"     "ΔbosR _in_vitro_1" "ΔbosR _in_vitro_2"
## [13] "ΔbosR _in_vitro_3" "ΔbosR _in_vitro_4"

WT_vs_bosR_in_vitro %>%
  ggplot(aes(x = log2FoldChange)) +
  geom_histogram(bins = 100) +
  geom_vline(xintercept = 0, col = "red") +
  theme_bw() +
  labs(
    title = "Figure 5. WT vs ΔbosR In Vitro Fold Change Distribution",
```

```
    x = "Log2 Fold Change",
    y = "Gene Count")

## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

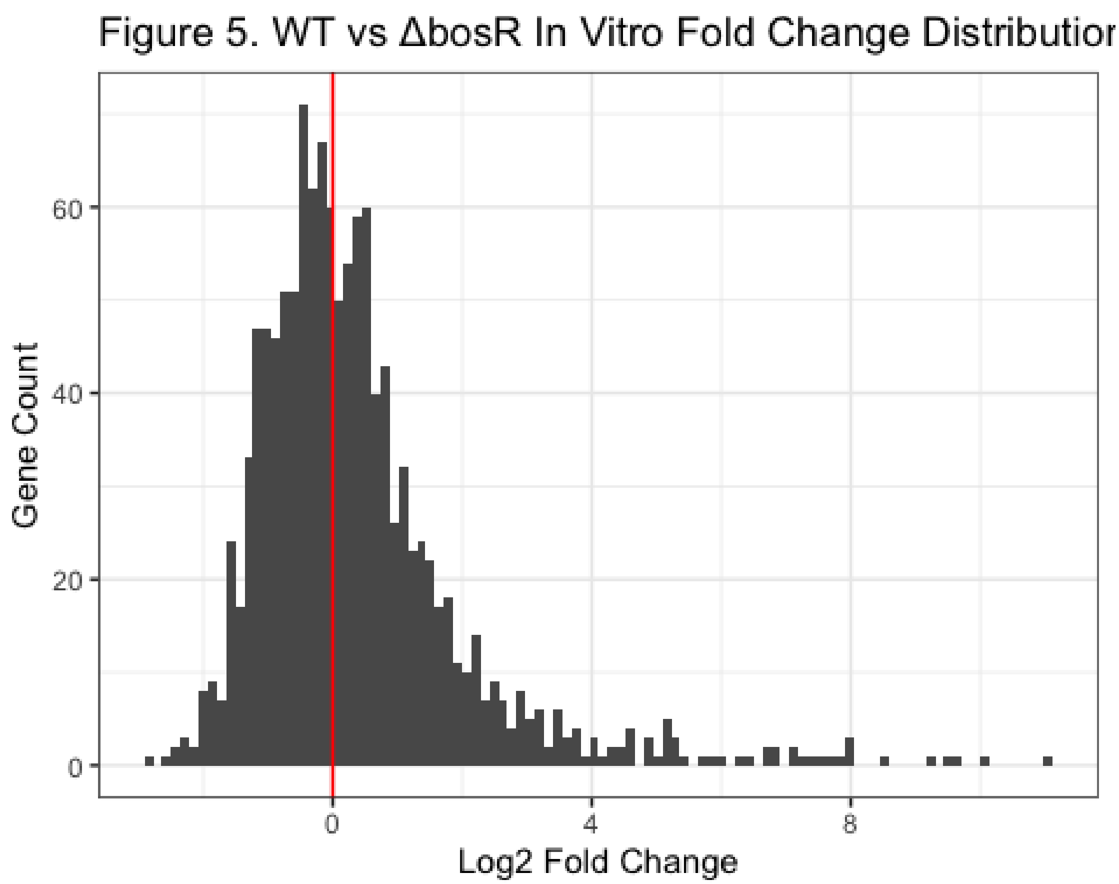

Figure 5. Histogram showing the distribution of log2 fold-change values between WT and ΔbosR strains under in vitro conditions. The distribution demonstrates widespread transcriptional changes associated with loss of BosR function.

```
WT_vs_bosR_in_vitro %>%
  ggplot(aes(log2FoldChange,padj)) +
  geom_point(shape = 1) +
  scale_y_log10() +
  geom_vline(xintercept = 0, linetype = 2) +
  geom_hline(yintercept = 1e-3, col = "red") +
  theme_bw() +
  labs(
    title = "Figure 6. WT vs ΔbosR In Vitro Volcano Plot",
    x = "Log2 Fold Change",
    y = "Adjusted p-value")

## Warning in scale_y_log10(): log-10 transformation introduced infinite
values.
```

```
## Warning: Removed 2 rows containing missing values or values outside the
scale range
## (`geom_point()`).
```

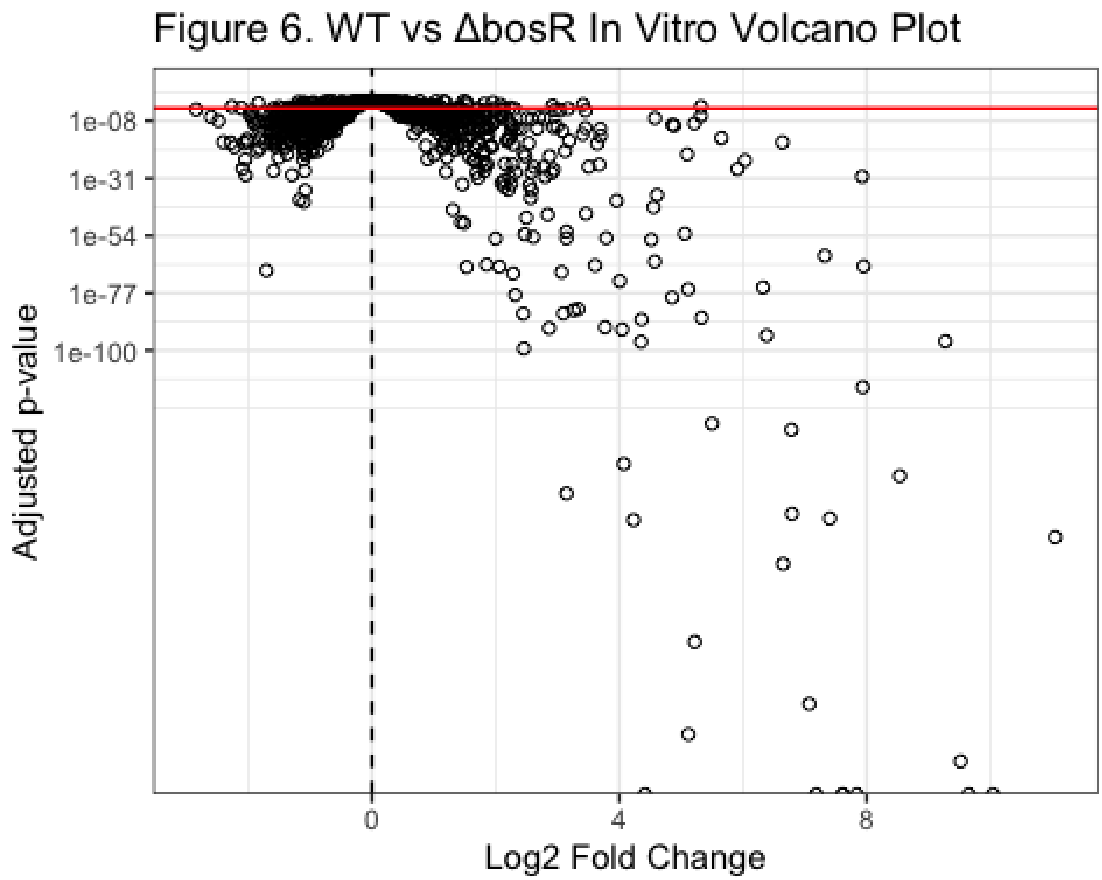

Figure 6. Volcano plot comparing WT and ΔbosR transcriptomes under in vitro conditions. Genes with significant differential expression are distributed away from the center and indicate BosR-dependent regulation of transcription.

```
WT_vs_bosR_R39A_in_vitro <- read_excel("mmi70036-sup-0002-tables2.xlsx",
sheet = 4)
colnames(WT_vs_bosR_R39A_in_vitro)

##  [1] "GeneID"                "baseMean"              "log2FoldChange"
##  [4] "lfcSE"                 "stat"                  "pvalue"
##  [7] "padj"                  "WT_in_vitro_1"         "WT_in_vitro_2"
## [10] "WT_in_vitro_3"         "bosR-R39A_in_vitro_1" "bosR-R39A_in_vitro_2"
## [13] "bosR-R39A_in_vitro_3" "bosR-R39A_in_vitro_4"

WT_vs_bosR_R39A_in_vitro %>%
  ggplot(aes(x = log2FoldChange)) +
  geom_histogram(bins = 100) +
  geom_vline(xintercept = 0, col = "red") +
  theme_bw() +
  labs(
```

```
    title = "Figure 7. WT vs BosR-R39A In Vitro Fold Change Distribution",
    x = "Log2 Fold Change",
    y = "Gene Count")
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

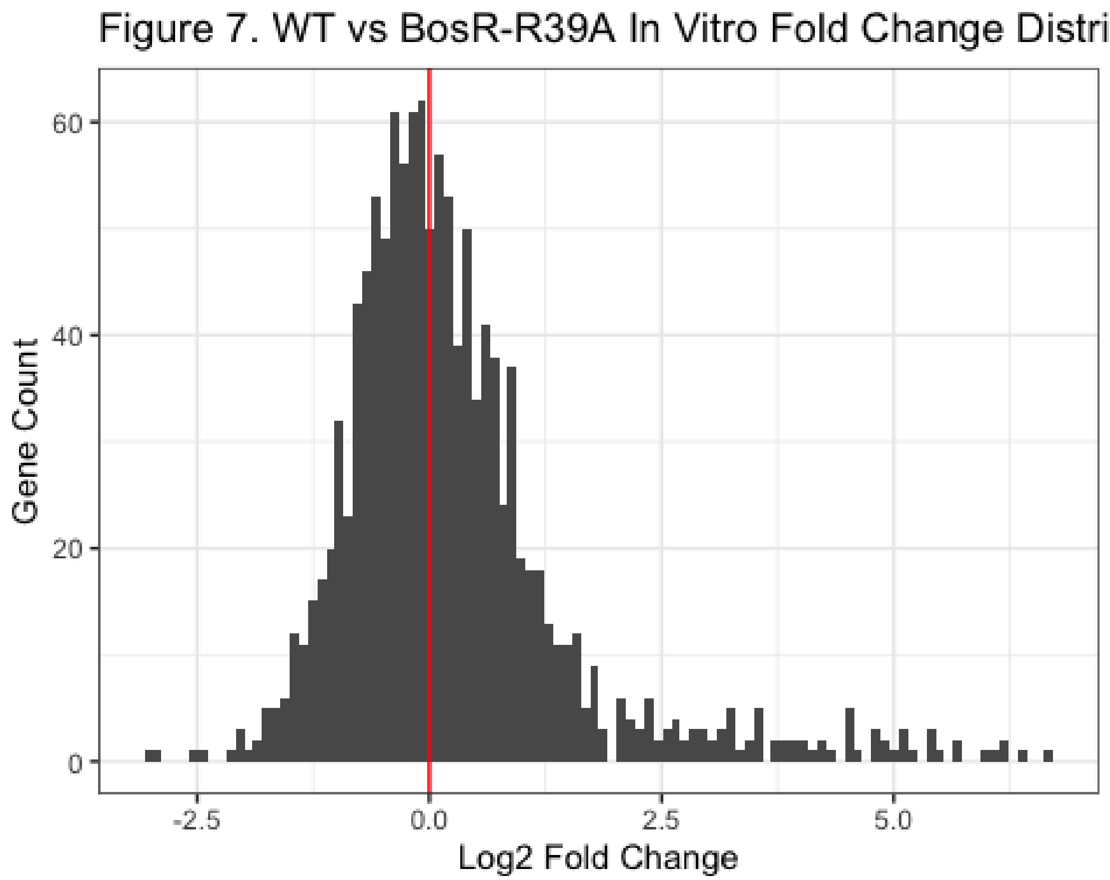

Figure 7. Histogram of log2 fold-change values comparing WT and BosR-R39A mutant strains under in vitro conditions. Mutation of the BosR DNA-binding residue altered expression of numerous genes across the transcriptome.

```
WT_vs_bosR_R39A_in_vitro %>%
  ggplot(aes(log2FoldChange, padj)) +
  geom_point(shape = 1) +
  scale_y_log10() +
  geom_vline(xintercept = 0, linetype = 2) +
  geom_hline(yintercept = 1e-3, col = "red") +
  theme_bw() +
  labs(
    title = "Figure 8. WT vs BosR-R39A In Vitro Volcano Plot",
    x = "Log2 Fold Change",
    y = "Adjusted p-value")
```

```
## Warning: Removed 2 rows containing missing values or values outside the
scale range
## (`geom_point()`).
```

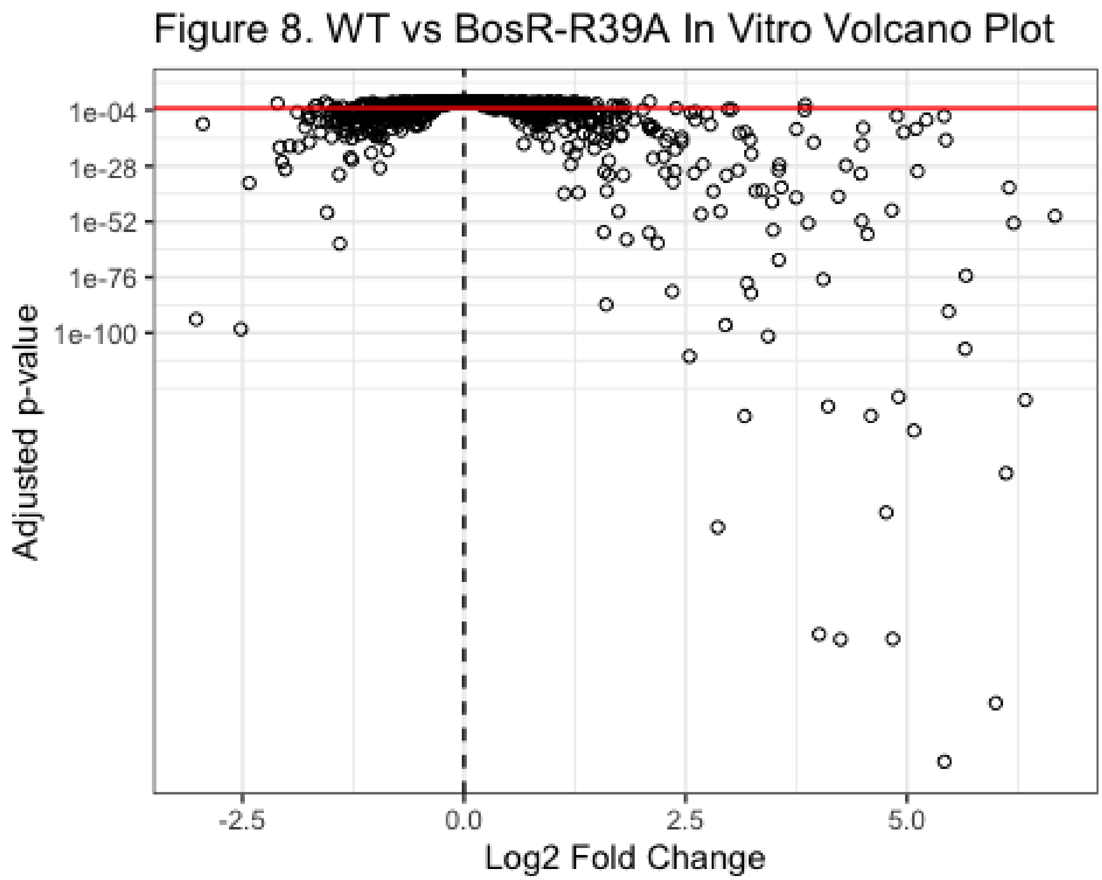

Figure 8. Volcano plot comparing WT and BosR-R39A mutant strains under in vitro conditions. Significant transcriptional differences demonstrate the importance of BosR DNA-binding activity in regulating gene expression.

table 3 DMC

```
WT_vs_bosR_DMC <- read_excel("mmi70036-sup-0003-tables3.xlsx", sheet = 3)
colnames(WT_vs_bosR_DMC)

##  [1] "GeneID"          "baseMean"        "log2FoldChange" "lfcSE"
##  [5] "stat"            "pvalue"          "padj"           "WT_DMC1"
##  [9] "WT_DMC2"         "WT_DMC3"         "WT_DMC4"        "WT_DMC5"
## [13] "WT_DMC6"         "ΔbosR _DMC1"     "ΔbosR _DMC2"    "ΔbosR _DMC3"
## [17] "ΔbosR _DMC4"     "ΔbosR _DMC5"

WT_vs_bosR_DMC %>%
  ggplot(aes(x = log2FoldChange)) +
  geom_histogram(bins = 100) +
  geom_vline(xintercept = 0, col = "red") +
```

```
  theme_bw() +
  labs(
    title = "Figure 9. WT vs ΔbosR DMC Fold Change Distribution",
    x = "Log2 Fold Change",
    y = "Gene Count")

## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_bin()`).
```

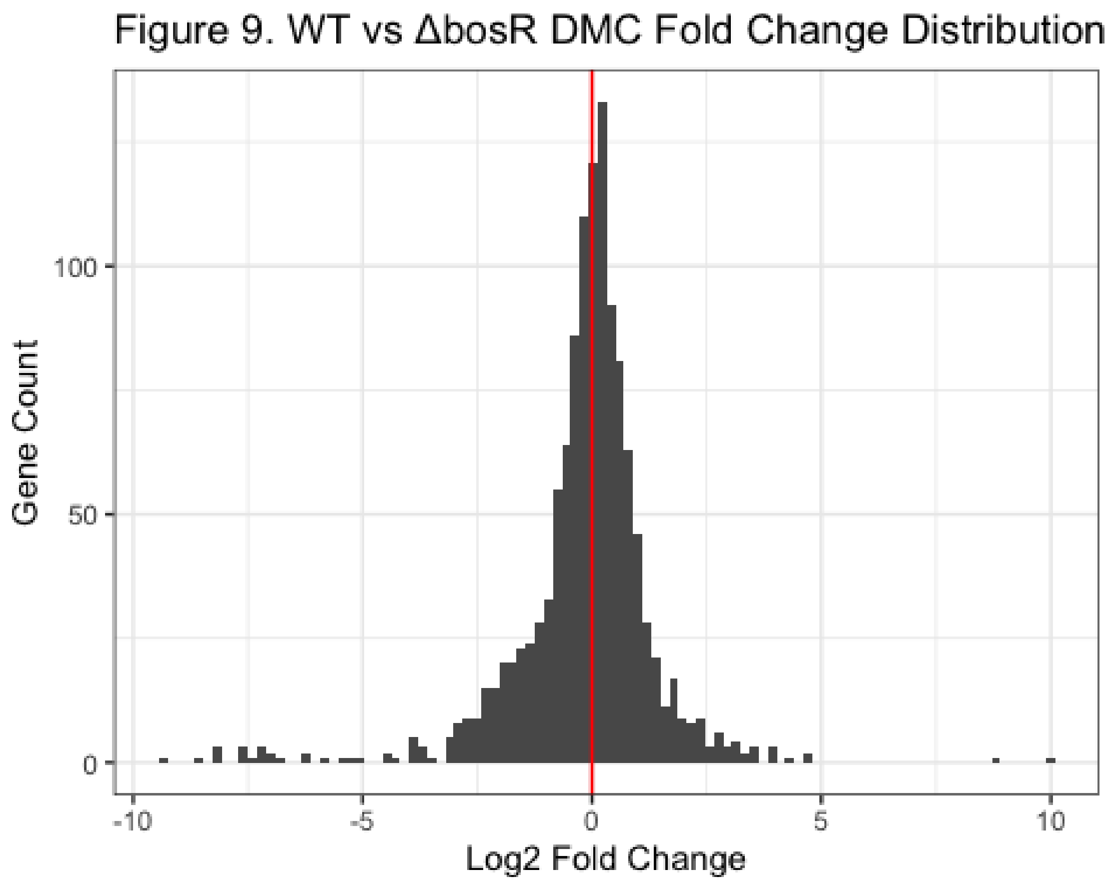

Figure 9. Histogram showing the distribution of log2 fold-change values between WT and ΔbosR strains under mammalian host-adapted DMC conditions. Broad transcriptional shifts indicate BosR-dependent regulation during host adaptation.

```
WT_vs_bosR_DMC %>%
  ggplot(aes(log2FoldChange, padj)) +
  geom_point(shape = 1) +
  scale_y_log10() +
  geom_vline(xintercept = 0, linetype = 2) +
  geom_hline(yintercept = 1e-3, col = "red") +
  theme_bw() +
  labs(
    title = "Figure 10. WT vs ΔbosR DMC Volcano Plot",
    x = "Log2 Fold Change",
    y = "Adjusted p-value")
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite
values.

## Warning: Removed 1 row containing missing values or values outside the
scale range
## (`geom_point()`).
```

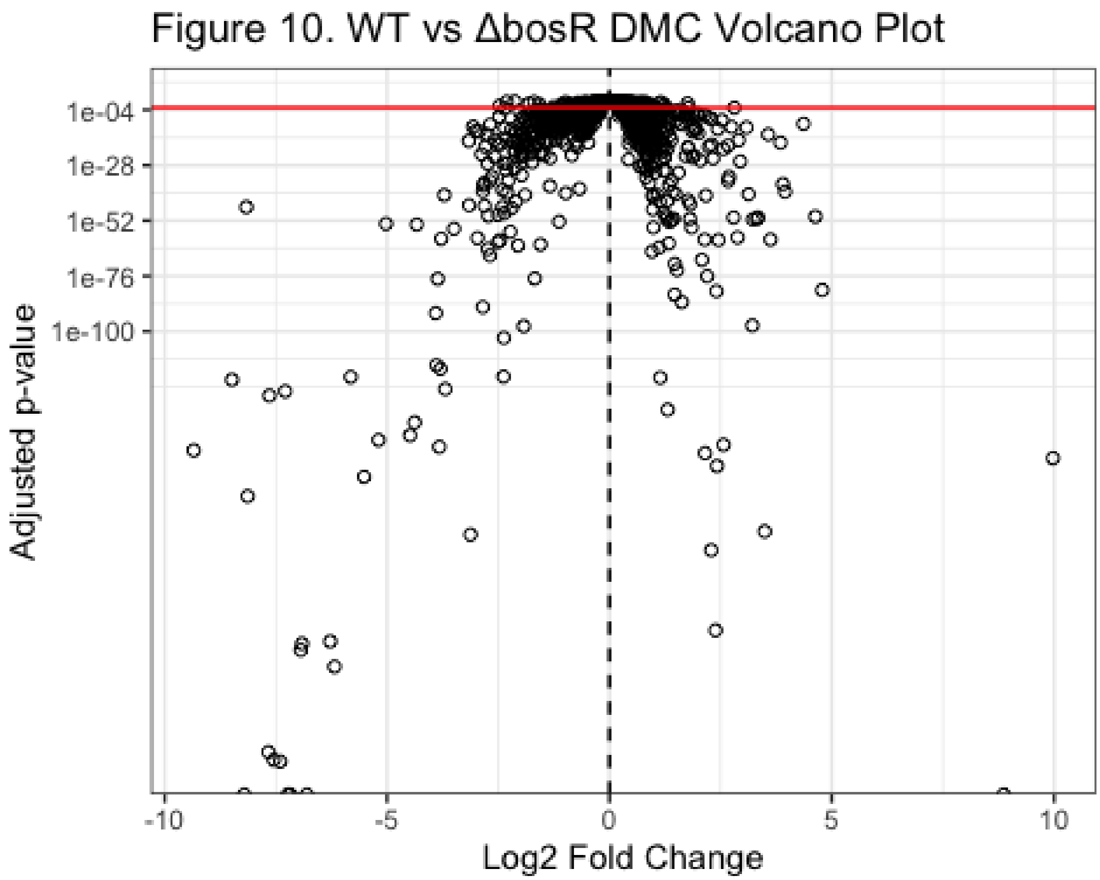

Figure 10. Volcano plot comparing WT and ΔbosR transcriptomes under mammalian host adapted DMC conditions. Differentially expressed genes indicate that BosR strongly regulates mammalian-phase gene expression and virulence associated pathways.

```
WT_vs_bosR_R39A_DMC <- read_excel("mmi70036-sup-0003-tables3.xlsx", sheet =
4)
colnames(WT_vs_bosR_R39A_DMC)

##  [1] "GeneID"           "baseMean"         "log2FoldChange"   "lfcSE"
##  [5] "stat"             "pvalue"           "padj"             "WT_DMC1"
##  [9] "WT_DMC2"          "WT_DMC3"          "WT_DMC4"          "WT_DMC5"
## [13] "WT_DMC6"          "bosR-R39A _DMC1" "bosR-R39A _DMC2" "bosR-R39A
_DMC3"
## [17] "bosR-R39A _DMC4"

WT_vs_bosR_R39A_DMC %>%
  ggplot(aes(x = log2FoldChange)) +
```

```r
  geom_histogram(bins = 100) +
  geom_vline(xintercept = 0, col = "red") +
  theme_bw() +
  labs(
    title = "Figure 11. WT vs BosR-R39A DMC Fold Change Distribution",
    x = "Log2 Fold Change",
    y = "Gene Count")

## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

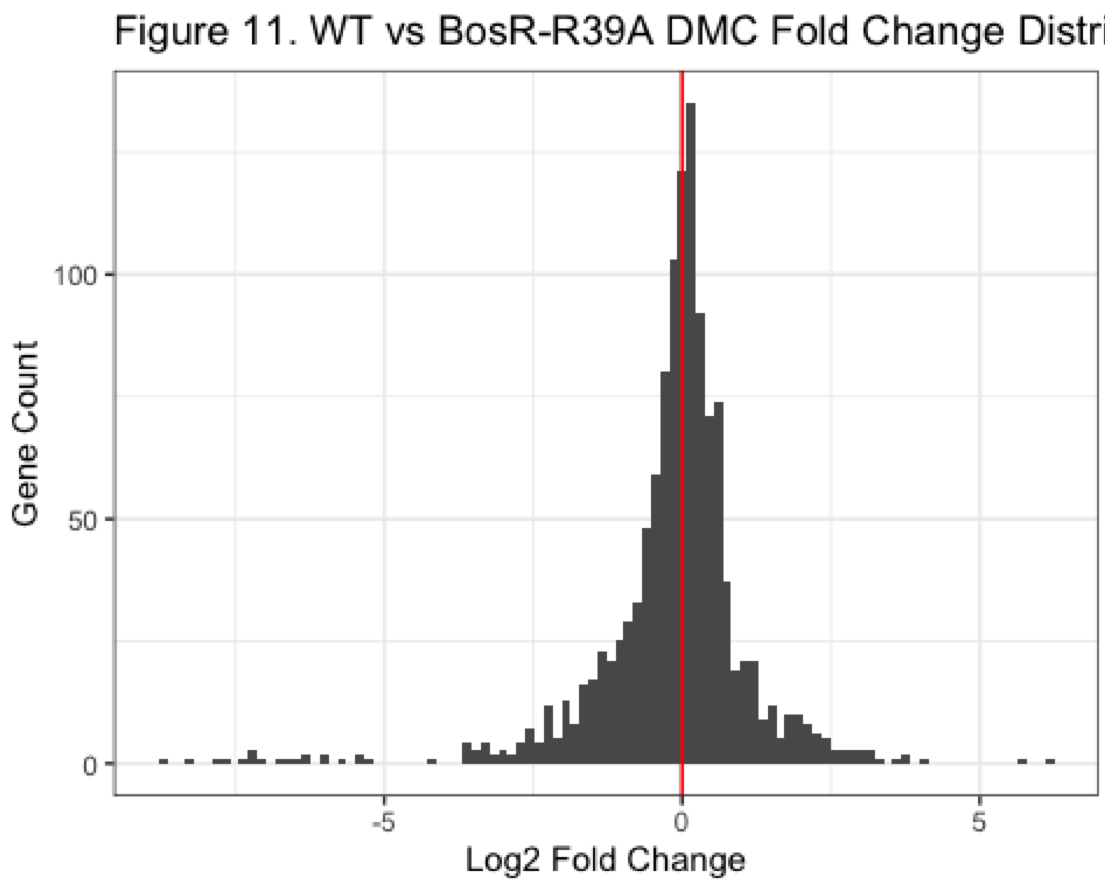

Figure 11. Histogram showing the distribution of log2 fold-change values between WT and BosR-R39A mutant strains under mammalian host-adapted DMC conditions. Mutation of the BosR DNA-binding residue resulted in widespread transcriptional changes during host adaptation.

```r
WT_vs_bosR_R39A_DMC %>%
  ggplot(aes(log2FoldChange, padj)) +
  geom_point(shape = 1) +
  scale_y_log10() +
  geom_vline(xintercept = 0, linetype = 2) +
  geom_hline(yintercept = 1e-3, col = "red") +
  theme_bw() +
  labs(
```

```
    title = "Figure 12. WT vs BosR-R39A DMC Volcano Plot",
    x = "Log2 Fold Change",
    y = "Adjusted p-value")
```

```
## Warning: Removed 2 rows containing missing values or values outside the
scale range
## (`geom_point()`).
```

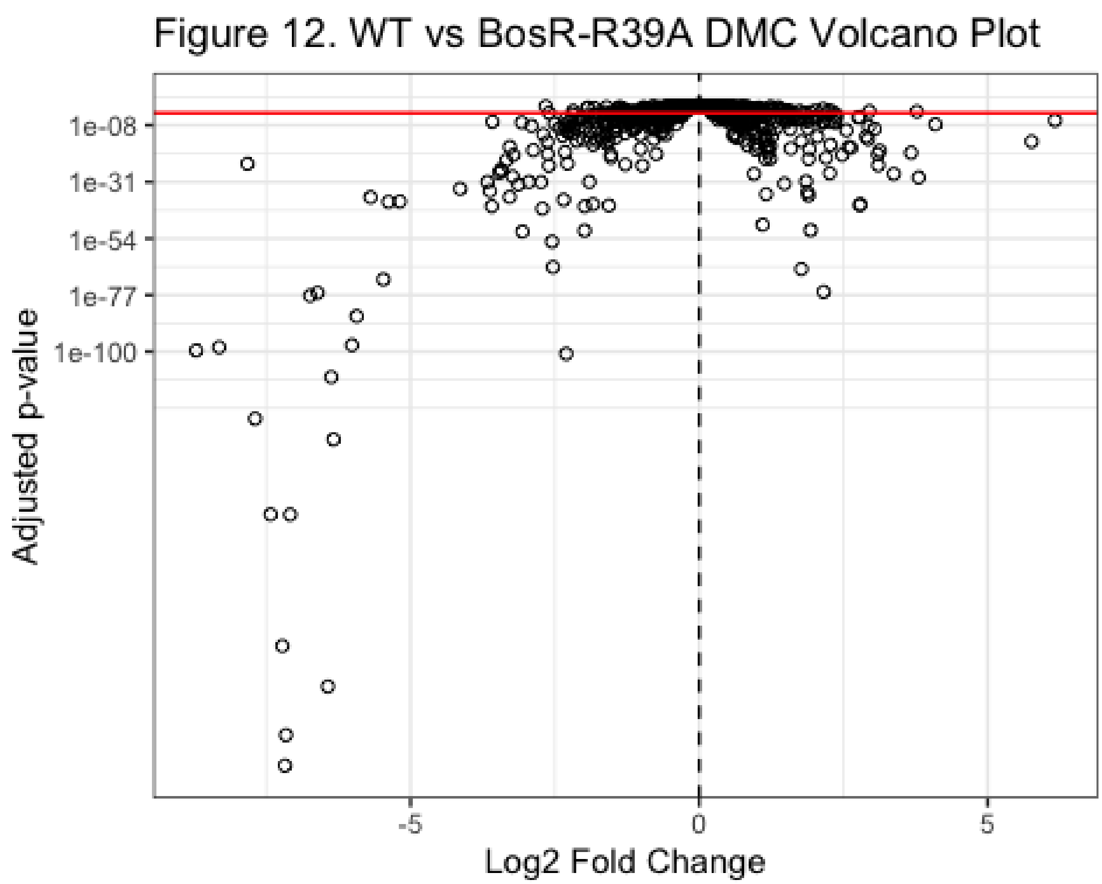

Figure 12. Volcano plot comparing WT and BosR-R39A mutant transcriptomes under mammalian host adapted DMC conditions. Significant differential expression demonstrates that BosR DNA-binding activity is essential for regulation of mammalian-phase and virulence associated genes.

4 Results

The exploratory analysis of the RNA-seq data allowed us to show the clear differences in transcriptomic profiles between Borrelia burgdorferi strains grown under in vitro and mammalian host-adapted DMC conditions. The histograms of normalized RFPKM expression values showed us the broad but consistent expression distributions across biological replicates after quantile normalization and log2 transformation which showed us how successful normalization was.

Comparison of WT in vitro and WT DMC samples revealed substantial transcriptional remodeling associated with mammalian host adaptation. The fold-change analysis allowed us to see the numerous genes that had a strong positive or negative log2 fold changes which allowed us to see how environmental conditions strongly influenced bacterial gene expression. The volcano plots allowed us to further demonstrate how many genes exhibited statistically significant differential expression following Benjamini–Hochberg correction with several genes surpassing the thresholds of adjusted $p < 0.05$ and $|\log2FC| > 1$.

The analysis of WT versus ΔbosR transcriptomes under in vitro conditions showed a widespread dysregulation of gene expression in the absence of BosR. The histograms of log2 fold-change values showed a broad distribution centered around zero, while the volcano plots showed the highly upregulated and downregulated genes in the ΔbosR mutant. Both of these findings allowed us to support the role of BosR as a global transcriptional regulator. Similar patterns were also observed in comparisons between WT and the BosR-R39A DNA-binding mutant which indicated that DNA-binding activity is very critical for proper BosR regulatory function.

The published RNA-seq analysis by Grassmann et al. (2025) further demonstrated that BosR regulates both RpoS-dependent and RpoS-independent genes under in vitro and DMC conditions. Principal component analysis (PCA) clearly separated the WT, ΔbosR, and ΔrpoS transcriptomes which proved distinct transcriptional states between the strains. The in vitro analysis identified 459 significantly dysregulated genes in the ΔbosR mutant which included many genes that were within the RpoS regulon. Under mammalian host adapted DMC conditions, BosR regulated 364 genes which included virulence associated genes such as ospC, dbpA, dbpB, and bbk32.

Genes that were previously associated with oxidative stress regulation like sodA, napA, and cdr were not significantly altered in the ΔbosR strain under the tested conditions. Instead, the BosR-dependent regulation was observed for genes involved in DNA repair, chemotaxis, chromosome maintenance, and host adaptation.

5 Conclusions

5.1 Biological conclusions

This study demonstrates that BosR functions as a major global transcriptional regulator in Borrelia burgdorferi and that it plays a central role in the bacterium's adaptation to mammalian host environments. Both of the loss of BosR or the disruption of its DNA-binding activity caused changes in gene expression which confirmed that BosR-mediated DNA binding is necessary for proper transcriptional regulation.

The results also support the conclusion that BosR regulates gene expression through both RpoS-dependent and RpoS-independent pathways. BosR affects the expression of virulence associated mammalian phase genes ospC and dbpA which are required for successful host infection and survival. At the same time, BosR independently regulates additional genes involved in chemotaxis, DNA repair, and cellular maintenance which suggests a broader regulatory functions that us beyond activation of the RpoS regulon.

These findings challenge the traditional idea that BosR primarily controls oxidative stress response genes since genes like sodA and napA were not significantly differentially expressed in the ΔbosR mutant under either in vitro or DMC conditions. This ultimately suggests that BosR's primary biological role is to coordinate transcriptional adaptation that are required for host colonization and virulence rather than directly regulating oxidative stress defenses.

Overall, the re-analysis allowed us to confirms that BosR is a multifunctional regulator that is very essential for mammalian host adaptation in B. burgdorferi. These findings allows us to improve our understanding of Lyme disease pathogenesis and provides us insight into how transcriptional regulation contributes to bacterial survival and infection within different host environments.

5.2 Future work

Future studies could be done to further investigate how BosR interacts with RNA polymerase and whether additional protein cofactors contribute to BosR-mediated transcriptional regulation.

Additional experiments could also determine:

- Whether BosR directly binds promoters of all identified target genes
- How environmental conditions influence BosR activity
- Whether BosR contributes to virulence during infection in animal models
- How BosR-dependent pathways differ between tick and mammalian host environments

Further characterization of BosR DNA-binding specificity may clarify why no strongly conserved DNA-binding motif was identified in this study.

6 References

6.1 Cite paper

Grassmann AA, McLain MA, Freeman MR, Caimano MJ, Radolf JD. DNA Binding by BosR Controls RpoS-Dependent and -Independent Gene Expression in Borrelia burgdorferi. Mol Microbiol. 2026 Feb;125(2):85-107. doi: 10.1111/mmi.70036. Epub 2025 Nov 14. PMID: 41239753; PMCID: PMC13036749.

6.2 Cite code repository

Computer codes & data: https://github.com/weigangq/MonteCarlo/tree/main/Bb-transcriptomes-Toka

6.3 Cite data file

Data set:

https://github.com/weigangq/MonteCarlo/blob/main/Bb-transcriptomes-Toka/mmi70036-sup-0003-tables3.xlsx

https://github.com/weigangq/MonteCarlo/blob/main/Bb-transcriptomes-Toka/mmi70036-sup-0002-tables2.xlsx